

AI Health Monitoring Guide

Project Report

Current Result

Risk Label

Medium

Health Risk Score

5

Comorbidity Count

2

Glucose: Normal | BP: Stage 2 | Cholesterol: Normal

[Explanation](#) [Weekly Trend](#) [Voice Summary](#)

Here is your health monitoring report:

1. Explanation of result

The predicted risk for today is **Medium**.

Let's look at your readings:

- **Blood Pressure:** Your systolic blood pressure is 153.6 mmHg and your diastolic blood pressure is 105.0 mmHg. Both of these readings are in **Stage 2 Hypertension**. This is a concerning finding, especially since you have existing Hypertension and Heart Disease. Compared to your previous 3-day average (148.8/90.1 mmHg) and 5-day average (148.3/90.6 mmHg), both your systolic and diastolic blood pressure are significantly higher today. Your blood pressure has been consistently elevated over the past several days, and today's reading indicates a worsening trend.
- **Blood Glucose:** Your blood glucose reading is 95.0 mg/dL, which is within the generally normal range. This is an improvement compared to your previous four days, which were all in the "elevated risk"

Ask about risk, diet, weekly trend, or next steps



Prepared for GenAI-LLM project submission

1. Abstract

AI Health Monitoring Guide is an AI-powered health monitoring assistant designed to interpret patient health readings more meaningfully than a simple threshold-based system. The project combines structured machine learning with GenAI-based interaction. The LightGBM model predicts a patient risk label using today's readings together with recent history, while the application layer explains the result, summarizes trends, and supports interactive follow-up questions. The goal of the system is to make daily monitoring easier for patients, caregivers, and evaluators by presenting predictions in clear and practical language.

2. Problem Statement and Need

Patients often record daily values such as blood glucose, blood pressure, pulse, and cholesterol, but these numbers are difficult to interpret in isolation. A single reading may not fully represent the patient's condition. In practice, health risk is often identified from patterns over multiple days, such as sustained high blood pressure, worsening glucose levels, or recurring abnormal readings. Many basic applications store data but do not provide usable interpretation. This project addresses that gap by combining prediction, explanation, and simple interaction in one interface.

This project can be very useful for all the senior living communities in India, we can integrate with them for monitoring senior's citizen's health conditions everyday

3. Objectives

- Predict a patient risk label using current values and recent historical context.
- Engineer medically meaningful features such as 3-day and 5-day averages, day-over-day change, and comorbidity count.
- Present the output in simple language instead of only showing raw scores.
- Provide weekly trend insight and voice-friendly summaries for better usability.
- Support a Q&A interface so users can ask follow-up questions about the result.

4. Dataset and Features

The project uses patient health records containing Patient ID, Date, Day, Age, Gender, BMI, Blood Glucose, Systolic BP, Diastolic BP, Pulse, Cholesterol, Diabetes, Hypertension, and Heart Disease. Where an explicit risk label is not available, the system derives a label through medical-guideline-based logic. This derived label helps train the machine learning model while preserving a clinically interpretable basis for risk categorization.

- Key engineered features include:
 - 3-day and 5-day rolling averages of glucose, blood pressure, pulse, and cholesterol
 - Difference between current readings and recent averages
 - Difference between current values and yesterday's values
 - Counts of high-risk days over the recent 5-day window
 - Blood pressure stage, glucose band, cholesterol band, rule-based risk score, and comorbidity count

5. System Design and Workflow

The end-to-end workflow of the system is summarized below.

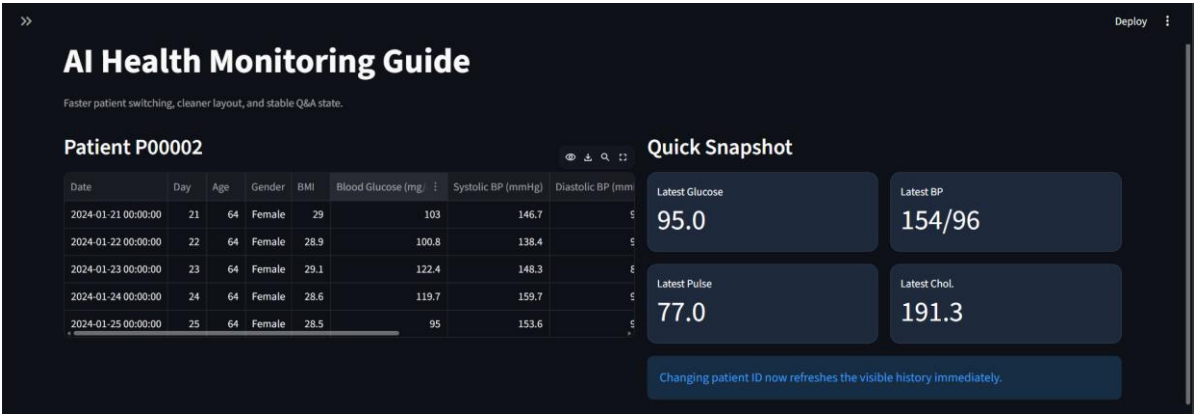
Stage	Description
Data loading	The patient dataset is loaded and cleaned. Records are sorted by Patient ID and Date.
Feature engineering	Medical-guideline features and rolling history features are created.
Model inference	LightGBM predicts the risk label from current values plus historical features.
Result interface	The application shows risk label, risk score, comorbidity count, and supporting explanation.
GenAI layer	Gemini is used for explanation, weekly trend insight, voice summary, and Q&A assistance when available.

6. Model Choice and Rationale

LightGBM was selected as the final model because the dataset is structured and tabular, and the prediction task depends on multiple engineered numerical features. LightGBM is efficient, fast, and well suited to this type of input. It handles non-linear relationships effectively and provides strong performance without requiring a large deep learning pipeline. The model is therefore a practical choice for a responsive health monitoring interface.

7. Application Interface

The Streamlit application is designed to make the workflow intuitive. A user first selects the patient ID, reviews the recent readings, enters today's values, and then generates the current prediction. The interface presents both structured metrics and explanatory text, making the system suitable for demonstration and evaluation.



Patient history and quick snapshot

Enter Today's Input

Day	Blood Glucose (mg/dL)	Cholesterol (mg/dL)
25	95.00	191.30
Age	Systolic BP (mmHg)	Diabetes
64	153.60	0
Gender	Diastolic BP (mmHg)	Hypertension
Female	105.00	1
BMI	Pulse (bpm)	Heart Disease
28.50	77.00	1

Predict Risk

Today's input form

Current Result

Risk Label
Medium

Health Risk Score
5

Comorbidity Count
2

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Ask about risk, diet, weekly trend, or next steps

Current result and explanation

Current Result

Risk Label
Medium

Health Risk Score
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Comorbidity Count
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Glucose: Normal | BP: Stage 2 | Cholesterol: Normal

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Hello! We have some good news today. Your blood glucose and cholesterol are looking healthy.

However, your blood pressure is still quite high, in the Stage 2 range. It's also shown a concerning upward trend this week.

Your overall risk is currently assessed as medium.

The most important thing to focus on now is bringing your blood pressure down. We'll continue to monitor your progress together.

0:00 / 0:29

Generate Voice Summary

Voice summary is generated only when you click the button, so the app feels faster.

Ask about risk, diet, weekly trend, or next steps

Voice summary interface

8. GenAI Components

- Explanation layer: converts the model output into simple health-oriented language.
- Weekly trend insight: summarizes improvement, deterioration, instability, and attention areas.
- Voice summary: creates a spoken version of the current result for easier listening.

- Q&A layer: allows follow-up questions about the predicted result and patient context.
- RAG layer has also been used for generating better explanations

These layers improve usability, especially for non-technical users. However, the system should still be described as an AI-powered health monitoring assistant rather than a fully agentic AI system, because it does not autonomously plan and act without user control.

9. Strengths, Limitations, and Future Scope

- Strengths
 - Combines machine learning and GenAI in a clear workflow.
 - Uses historical context instead of only single-day values.
 - Gives both structured output and natural-language interpretation.
 - Includes voice and Q&A features for a better demonstration experience.
- Limitations
 - The GenAI layer depends on API quota and connectivity.
 - This is a support system and not a substitute for medical advice or diagnosis.
- Future scope
 - Add multilingual voice output and multilingual Q&A.
 - Create doctor/caregiver reports automatically.
 - Improve offline fallback logic when API services are unavailable.
 - Add scheduling, reminders, and more autonomous workflow handling.

10. Conclusion

AI Health Monitoring Guide demonstrates how a structured ML model and GenAI features can work together in a practical health-monitoring application. LightGBM provides the predictive backbone, while the explanation, weekly insight, voice summary, and Q&A layers make the output more useful and understandable. The project is therefore a strong academic demonstration of an AI-powered health monitoring assistant built on tabular healthcare-style data and user-friendly application design.

11. Project Links

Replace the placeholders below before final submission:

- Live Demo: <https://ageless-abiding-related.ngrok-free.dev>
- Code Repository: [Lohith284/AI-HEALTH-MONITORING-GUIDE](https://github.com/Lohith284/AI-HEALTH-MONITORING-GUIDE)